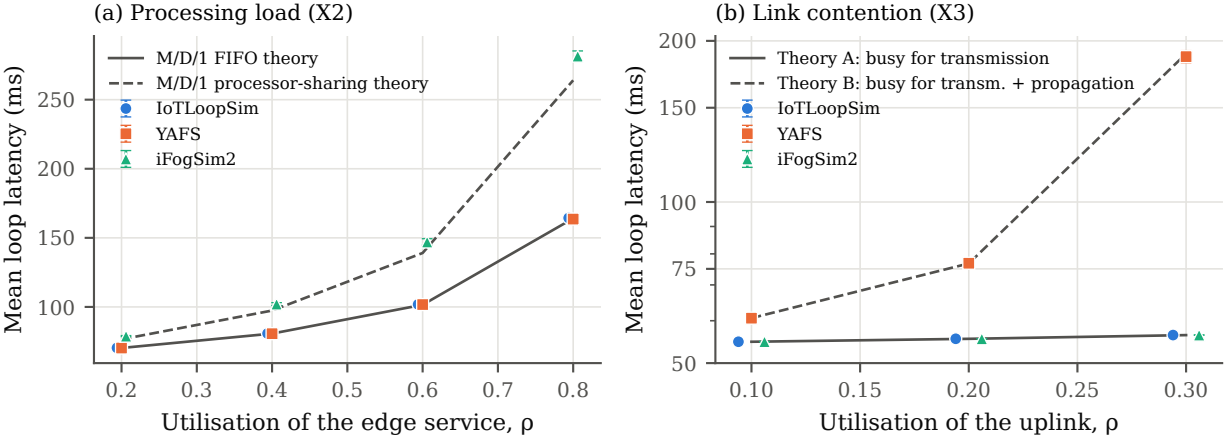
(a) Processing load (X2)
M/D/1 FIFO theory
M/D/1 processor-sharing theory
IoTLoopSim
YAFS
iFogSim2
Mean loop latency (ms)
100
150
200
250
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Utilisation of the edge service, ρ
(b) Link contention (X3)
Theory A: busy for transmission
Theory B: busy for transm. + propagation
IoTLoopSim
YAFS
iFogSim2
Mean loop latency (ms)
50
75
100
150
200
0.10
0.15
0.20
0.25
0.30
Utilisation of the uplink, ρ